

Groups of Permutations

Section 8

Def: A permutation of a set A is a bijection from A to A .

$$S_A := \{ \sigma \mid \sigma \text{ is a permutation of } A \}$$

Ex: $A = \{1, 2, 3, 4, 5\}$, $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix}$

$$\sigma(1) = 4$$

$$\sigma(5) = 1$$

$$\sigma(2) = 2$$

$$\sigma(3) = 5$$

$$\sigma(4) = 3$$

two-line notation

Prop: S_A is closed under composition.

Pf: We need to check that if $\sigma, \pi \in S_A$ then $\sigma \circ \pi : A \rightarrow A$ is a bijection.

① the composite of injective functions is injective

② the composite of surjective functions is surjective.



Ex: $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix}$ $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix}$

$$\sigma \tau = \sigma \circ \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 3 & 2 & 4 \end{pmatrix}$$

↑
permutation
multiplication

$$\tau \sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 1 & 4 & 3 \end{pmatrix}$$

↑
not equal

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 4 & 1 & 3 \end{pmatrix}$$

Thm: Let A be any set. Then S_A is a group under permutation multiplication.

Pf:

① Associative: functional composition is associative ✓

② Identity: $e: A \rightarrow A$ is defined by $e(a) = a, \forall a \in A$. ✓

③ Inverse: A bijection $\overset{S_A}{\sigma}: A \rightarrow A$ has an inverse function $\sigma^{-1}: A \rightarrow A$ which is also a bijection. And $\sigma \circ \sigma^{-1} = e$ so σ^{-1} is the inverse of σ in S_A . $\square$

For $A = \{1, 2, \dots, n\}$ we write S_n for S_A . The symmetric group on n letters.

Note: $|S_n| = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1 = n!$

We will usually consider A finite.

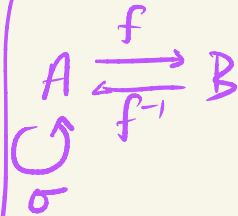
Prop: If $|A| = |B|$ then $S_A \cong S_B$.

Pf: Let $f: A \rightarrow B$ be a bijection. For $\sigma \in S_A$ let's define $\phi(\sigma) = f \circ \sigma \circ f^{-1} \in S_B$. Then

$\phi: S_A \rightarrow S_B$ is a group isomorphism.

Homomorphism: $\phi(\sigma \circ \pi) = f \circ \sigma \circ \pi \circ f^{-1} = (f \circ \sigma \circ f^{-1}) \circ (f \circ \pi \circ f^{-1})$

Bijection (Exercise) $= \phi(\sigma) \circ \phi(\pi)$ ✓



Example: S_1 $1! = 1$

$$\stackrel{1!}{=} \{e\}$$

Example: S_2 $2! = 2 \cdot 1 = 2$

$$\stackrel{2!}{=} \mathbb{Z}_2$$

$$S_2 = \left\{ \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \right\}$$

$$\stackrel{1!}{=} e$$

Example: S_3 . $\sigma_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$

- not abelian

$$\sigma_1 \sigma_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\sigma_2 \sigma_1 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

S_3 is the smallest
nonabelian group

- $\langle \sigma_1, \sigma_2 \rangle = S_3$

$$S_3 = \{ e, \sigma_1, \sigma_2, \sigma_1 \sigma_2, \sigma_2 \sigma_1, \sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2 \}$$
$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

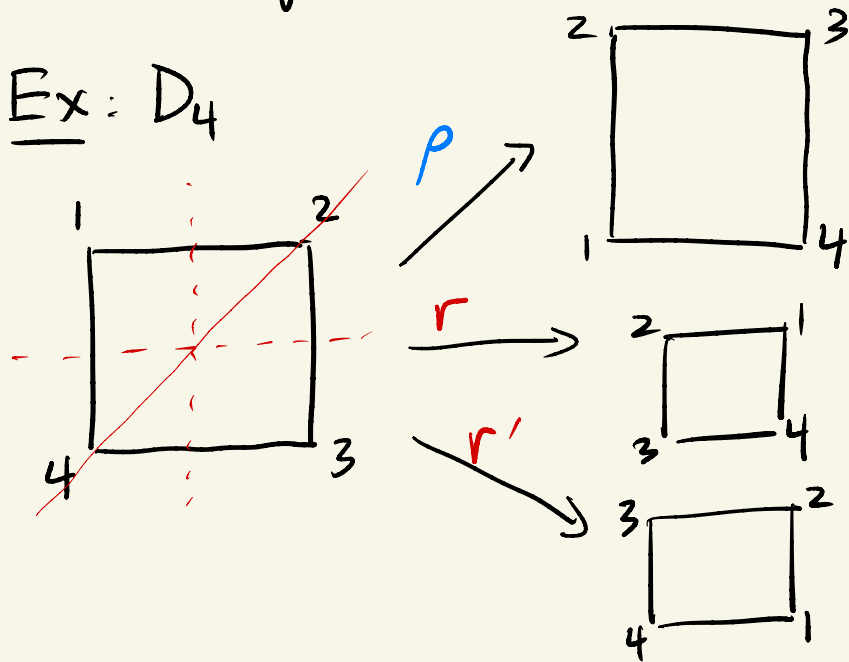
Dihedral Groups

Def: The **dihedral group** D_n is the group of symmetries of a regular n -gon.

$$|D_n| = 2n$$

$$, p^2, p^3, p^4 = e$$

Ex: D_4



reflect across vertical

reflect across diagonal

90° cclk
wise
rotation
↓
 $r' = p^{-1}r$

By viewing these as permutations of the vertices,
we get an isomorphism D_n to a subgroup of S_n .

In fact every group G has this property.

Thm (Cayley's Thm). Every group G is isomorphic
to a subgroup of S_G .

the group of permutations of
 G

Def: For $f: A \rightarrow B$ a function and $H \subseteq A$ define
 $f[H] = \{f(h) \mid h \in H\} \subseteq B$ the **image** of H under f .

Prop: Let G, G' be groups and $\phi: G \rightarrow G'$ a
homomorphism. Then $\phi[G] \subseteq G'$. If, moreover,
 ϕ is injective, then $\phi: G \rightarrow \phi[G]$ is an isomorphism.

Pf:
Satisfies homomorphism property
① **Closed**: let $x', y' \in \phi[G] \subseteq G'$. Then $\exists x, y \in G$
with $\phi(x) = x'$ and $\phi(y) = y'$. Now $x'y' = \phi(x)\phi(y) = \phi(xy) \in \phi[G]$.
② **Identity**: $e' = \phi(e) \in \phi[G]$
 $\uparrow$
 identity of G' .

③ Inverses: $\nexists x' \in \phi[G]$, so $\exists x \in G$ with $\phi(x) = x'$.

$$\text{Then } \phi(x^{-1})x' = \phi(x^{-1})\phi(x) = \phi(x^{-1}x) = \phi(e) = e'$$

so $\phi(x^{-1})$ is the inverse of x' .

$\uparrow$

$$\phi[G]$$

$$\text{So } \phi[G] \leq G'.$$

$\nexists$ that ϕ is injective. Consider $\phi: G \rightarrow \phi[G]$.

① ϕ is homomorphism: ✓

② Injective by assumption

③ Surjective by definition of $\phi[G]$. Thus $\phi[G] \cong G$. $\square$

Pf of Cayley's Thm: By the Proposition, it suffices to give a injective homomorphism $\phi: G \rightarrow S_G$.

Define $\phi(g)$ to be the function from $G \rightarrow G$ given by

$$\phi(g)(x) = gx.$$

$$\textcircled{1} \phi(g) \in S_G: (\phi(g) \circ \phi(g^{-1}))(x) = \phi(g)(g^{-1}x) = gg^{-1}x = x$$

hence $\phi(g^{-1})$ is an inverse function to $\phi(g)$, so $\phi(g) \in S_G$.

$$\textcircled{2} \phi \text{ is a homomorphism: } \phi(gh)(x) \stackrel{\text{WTS}}{=} \phi(g) \circ \phi(h)(x)$$

$$\begin{array}{ccc} \parallel & & \parallel \\ (gh)x & & \phi(g)(hx) = ghx \end{array}$$

$$\textcircled{3} \phi \text{ injective: } \phi(g)(e) = g \quad \text{so } \phi(g) = \phi(h) \implies g = h \quad \square$$